

Delete the rich

On the non-existence of ‘weak’ schwa deletion

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Abstract

It is widely assumed that a ‘weak’ schwa, a vowel with a minimal amount of featural specification, is often phonologically deleted across languages. There are, however, conceptual problems with this perspective, and under fairly standard assumptions, it turns out to be logically impossible to delete weak schwa to the exclusion of more highly specified vowels. In this light, the lack of robust empirical evidence for weak schwa deletion is not surprising. Given basic feature logic, richly specified segments can be deleted to the exclusion of more weakly specified segments. However, such underspecified segments can be *inserted* to the exclusion of more highly specified ones.

Keywords: *phonology, underspecification, insertion, deletion, schwa*

Obviously there is no way that the Air Force or anybody else can prove that alien spacecraft are not visiting us. Is there a tooth fairy? No amount of cases in which a grownup is caught pushing a quarter under a child's pillow will add up to irrefutable negative evidence. Always there is a small residue of cases in which grownups are not caught, and the morning appearance of money remains mysterious. No matter how many sightings of UFOs are shown to have natural explanations, there is always a residue—how could it be otherwise?—of cases for which information is insufficient for judgment.

Martin Gardner (1978), 'The Third Coming'

1 Introduction

Youngberg (2023) provides an overview of research positing a connection between *emptiness*, a lack of phonological specification (features, particles or elements), and schwa vowels. The work serves as an introduction to Part 3 'Emptiness, schwa, and epenthesis' of a volume (Breit et al. 2023) dedicated to Government Phonology and related approaches, and it also offers historical notes on the longstanding idea of a connection between representational emptiness (total or relative) and alternation of a schwa vowel with zero. Such alternations can be conceived of as reflecting insertion or deletion rules, or as realization or failure to be realized, depending upon one's theoretical commitments. Youngberg (p. 155) cites Anderson (1982) as an early application of the idea that schwa-like behavior reflects a lack of representational content: "Anderson (1982) proposed that schwa

is an empty nucleus which is [sometimes] realised as [œ].”

Anderson suggests that the French schwa contains no features whatsoever underlyingly, but that the features of a surface vowel [œ] are provided by rule when the vowel is realized. This idea, that a lack of representational content plays a role in explaining alternations of the appearance of a segment vs. the absence of a segment is a persistent notion in the field. This notion transcends differences among competing representational frameworks such as Government Phonology, Element Theory and the use of binary features as in *SPE* (Chomsky and Halle 1968) and related work like Anderson’s paper, and more recent work discussed below, including Flemming (2009) and Silverman (2011).

In the next section, I provide a sketch of some notational and feature logic problems with this idea, focussing on Anderson’s rules and representations. I then offer some notational and feature logic innovations in the context of a binary feature system. To anticipate my conclusion, the idea that a lack of features is equivalent to the absence of a segment will be critiqued as an error of set theoretic types. Each segment will be analyzed as composed of a set of features; since the empty set is not the same thing as nothing—the empty set is a set, just like an empty box is a box—the absence of features is not the same as the absence of a segment. By exposing problems with Anderson’s reasoning and use of notation, and by offering an alternative, I present a logically sound interpretation of phonological representation which plays a role in developing a coherent theory of what is a possible rule. At the end of the paper, we will see that the logical results appear to comport with the findings of several recent empirical studies of schwa-zero alternations. The conclusion is that it is logically impossible that weak, underspecified schwa be targeted for deletion, but that it can be the outcome of an insertion rule. There are also clear implications for other rules involving underspecified segments.

2 Formal and informal rule notations

Phonological rules are often presented in a mix of formats with various levels of formalization reflecting both expository convenience and current levels of understanding concerning different phenomena. For example, one might use a formula like ‘ $t \rightarrow d$ in Coda’, if the details of syllabification theory are not particularly important for the matter at hand. In *SPE* and related work the targets of rules are sometimes expressed as feature matrices. For example, a rule that targets the feature matrix in (1a) targets the natural

class of high front vowels in a language.

(1) Targets

$$\text{a. } \begin{bmatrix} +\text{HIGH} \\ -\text{BACK} \end{bmatrix} \quad \text{b. } [+\text{HIGH}] \quad \text{c. } \begin{bmatrix} +\text{HIGH} \\ -\text{BACK} \\ +\text{RD} \end{bmatrix}$$

By specifying fewer features, as in (1b), the target class is expanded to all high vowels. And by specifying more features, as in (1c), the target class is restricted to just high front round vowels. For lack of a better name, I call this inverse relationship between the number of features used in a partial description to intensionally define a natural class, and the number of elements in the extension of the class, the “tall, dark and handsome” principle (TDH)—the set of tall people is obviously a superset of the set of tall, dark, handsome people, but the latter uses a superset of features of the former.

Rules are something expressed informally with segment symbols instead of feature matrices, as in (2), but *SPE*¹ and work like Kenstowicz and Kisseberth (1979) make it clear that such symbols are just abbreviations for feature matrices.

(2) $y \rightarrow i$ in a closed syllable

In order to have a consistent model of what is a possible phonological rule, it is necessary to have a consistent interpretation of the components of rules. Therefore, it is necessary to interpret (2) not as a statement of what happens to *the segment* /y/, but rather what happens to *the natural class whose only member is the segment* /y/. These are different formal objects, just as the number 1 is different from the set {1}, the set whose only member is the number 1. In other words, assuming that the only high, front, round vowel in a language is /y/, the target of (2) should be understood as in (3) (abstracting away from other potentially relevant features).

$$(3) \begin{bmatrix} +\text{HIGH} \\ -\text{BACK} \\ +\text{RD} \end{bmatrix} \rightarrow i \text{ in a closed syllable}$$

This distinction between targeting individual segments (which just reflects an informal use of segment notation) and targeting natural classes whose extension is a singleton set containing one segment is typically ignored, but in one particular situation the difference is crucial.

¹E.g., on page 5: “The phonetic symbols [p], [t], [e], [i], [u], etc., are simply informal abbreviations for certain feature complexes”.

For a representative example of how a lack of precision in notation can lead to problems, consider the analysis of French schwa presented by Anderson (1982) which argues clearly for an underlying contrast between the vowels informally denoted /œ, ə/. Anderson's analysis is that the second of these, the so-called French schwa, which either surfaces as [œ] or is not pronounced at all, is "a structurally-present syllabic nucleus WITH NO ASSOCIATED PHONOLOGICAL FEATURES", a representation that allows us to "treat schwa as a sort of underlying /∅/" (p. 551). So, Anderson uses the empty set symbol to represent a lack of content for French schwa.

Anderson proceeds to suggest a number of rules whose targets are this segment /∅/. However, applying the same reasoning that we used to distinguish the segment /y/ from the singleton class that contains just the segment /y/, the featural representation of this natural class should be an empty pair of square brackets: []. By TDH, this does not correspond to a single segment, but rather to the set of *all* segments. Contrary to Anderson's title, you don't "get something for nothing" in a partial description, but rather you get *everything for nothing*. A partial description that is minimally specified defines a natural class whose extension is maximal. If (1a) defines the class of high front vowels, and (1c) defines the class of high front round vowels (potentially just /y/), then [] defines the class of all vowels (or maybe all segments).

We can now see that all of Anderson's rules that have as their target /∅/ do not do what he intended. Basic feature logic dictates that his rule (6), the target and change of which are formulated as in (4), cannot have his intended interpretation.

$$(4) \quad \emptyset \rightarrow [-\text{BACK}, -\text{HIGH}, -\text{LOW}, +\text{ROUND}]$$

Instead, the version using consistent notation is in (5).

$$(5) \quad [] \rightarrow [-\text{BACK}, -\text{HIGH}, -\text{LOW}, +\text{ROUND}]$$

And as we have seen, under a coherent rule formalism, this has to mean that the change applies to all vowels (or all segments), and not just schwa.

A further problem of Anderson's notation, is the double use of '∅' to represent the featural content of a fully unspecified segment and also the null segment that is the target of segment insertion rules and the structural change of segment deletion rules. In Anderson's notation, a rule of schwa deletion, assuming schwa is completely unspecified, would be the formula in (6).

$$(6) \quad \emptyset \rightarrow \emptyset$$

In fact, Anderson does provide a rule of, basically, this form, his Rule (9) presented in (7), along with the assumption that there is “a rule of French [which] has the property of deleting entirely any syllable which, at an appropriate level of a derivation, contains no specified material whatever.”²

$$(7) \quad [_{\sigma} \emptyset] \rightarrow \emptyset$$

The two tokens of ‘ $\emptyset$ ’ in (7) have to have completely different interpretations. For Anderson, the first means “a segment without features” and the second means “the absence of a segment”. The problem here is clear: just as ‘{1}’ is a different thing from ‘1’, so is the set that contains nothing, the empty set, which can be denoted by ‘{ }’ or $\emptyset$, different from nothing—the former, the empty set, is a set, but the latter, nothing, is nothing.

Before proceeding, we should note that an expression like (6), using Anderson’s notation, can also mean “insert schwa”. Clearly a notation that collapses “delete schwa” and “insert schwa” is problematic.

In the rest of this paper, I flesh out these logical issues discussed with regard to Anderson’s notation, because they are in fact widespread and confusing. I introduce a notation that is consistent in terms of set theory and the treatment of segments versus natural classes. I also introduce a distinct symbol for the null segment of insertion and deletion rules. Once the logic is explicit, we can derive a result that is perhaps unexpected: schwa, or any other segment that is underspecified relative to another segment in a language, cannot be targeted alone in the intensional characterization of a rule, whether the rule be one that changes the feature content of segments or one that deletes segments. In contrast, weak schwa can be the structural change of a segment *insertion* rule to the exclusion of other segments. The logical result, that ‘weak’ schwa deletion is logically impossible, but weak schwa insertion is possible, turns out to be fairly plausible in light of empirical data.

Phonological grammars are formal systems, so it is important to avoid in our grammatical models sloppy expressions that are intended to express rules that are logically inconsistent with basic mathematical notions of set theory. Another reason to clarify these issues is to show that widespread intuitions about the phonetic motivation for certain phonological processes turn out to be problematic, so this work is also a contribution to the idea of Substance Free Phonology, in the sense of Hale and Reiss (2008), Reiss (2017), Volenec and Reiss (2020). I hope to show a convergence of formal

²The square brackets in this rule mark the syllable edges.

logic, conceptualization and empirical data supporting the claim that a natural class consisting only of the weakly specified schwa vowel cannot be the intensionally defined target of a rule.

3 Thinking about mental representations

The bulk of this paper develops a formalism for segments and rules that allows us to challenge the still widespread assumption (e.g., van Oostendorp 2003, Flemming 2009, Silverman 2011), that the phonological behaviour of some so-called schwa vowels that alternate with zero reflects a representational poverty corresponding to extreme underspecification. In particular, I reject the idea that there are vowels, ‘weak schwas’, with an empty or minimal set of features, that occur in the lexicon of many languages, and which are commonly the target of syncope rules *because* their lack of specification means that they are already ‘weak’ and thus susceptible to deletion. I suggest that this perspective is a transform of “the mistake of attributing to a mental representation the properties of what it represents” (Pylyshyn 2003). The idea that this vowel disappears where other segments persist has been naively taken, by the whole phonological community it seems, as following from the notion that it doesn’t have as much content in its mental representation to begin with.

Of course, there is a lot of level-mixing here—a vowel is not a sound, but rather just a mental representation itself.³ Putting that aside, the important

³This point about linguistic categories is forgotten often enough that some scholars find it necessary to remind us on occasion, as illustrated by Chomsky and Halle (1968:ch. 7) and the following:

- “It should be sufficiently clear from this one example...how little the notion of speech sound is explicable in simple sensorimotor terms” (Sapir 1925).
- “The whole aim and spirit of this paper has been to show that phonetic phenomena are not physical phenomena *per se*, however necessary in the preliminary stages of inductive linguistic research it may be to get at the phonetic facts by way of their physical embodiment” (Sapir 1925).
- “It should be perfectly obvious by now that segments do not exist outside the human mind” (Hammarberg 1976).
- “Language, as far as I can tell, is *all* construction. ...[I]t begins to be difficult to see how the predicates *true* and *false* apply to one’s phonological representations in response to an incoming stimulus. ...The problems of reality get even worse when we think about syntactic structure. Is a syntactic structure real, something in the world?” (Jackendoff 1992:p. 164-5).

thing is that the mental representation of an elephant does not have to be bigger than that of a mouse, and that a mental representation with lots of parts doesn't necessarily corresponds to a thing in the world that is either bigger or more complex than a thing that the mind represents sparsely. My mental representation of the CERN collider is fairly simple, for example, compared to that of the surface of my desk—it is not clear which one I am more likely to change or forget. My mental representation of walking is less complex, I imagine, than my mental representation of dancing the macarena, but I am not therefore expected to stop walking more often than I stop dancing the macarena, or to lose the walking representation before the macarena representation as my brain degenerates.

There is just no clear line of reasoning that leads to the conclusion that simpler representations are likely to undergo deletion processes in phonology or any other domain. Perhaps it is useful to consider the syntax of ellipsis: it does not seem to be the case that VP-ellipsis is more likely to apply in cases like *John bought a coat and Mary did, too* than in *John bought a wool coat with fur trim and wooden buttons and Mary did, too*, although the elided VP of the first sentence contains a subset of the information in the elided VP of the second.

To complement this conceptual critique of common modeling of 'weak' segment behavior, I demonstrate that under fairly standard phonological assumptions, it is a *logical impossibility* that segment deletion rules target relatively underspecified segments to the exclusion of more specified ones. In contrast, segment insertion rules can *insert* underspecified elements.

Topping off the conceptual and logical arguments, I provide a brief discussion showing that the empirical evidence for deletion of 'weak' schwa is not at all robust. Of course, it is impossible to demonstrate a negative, the claim that weak schwa is never deleted, but it is surprising how weak the evidence for weak schwa deletion is. In fact, some of the most careful work on schwa-zero alternations, such as Vaux (1998) and van Oostendorp (1995) is consistent with the conclusion that epenthesis-based analyses fare better for many phenomena previously understood as reflecting syncope.

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- "it is idle to seek a mind-independent construct that corresponds to the syllable [ba]" (Chomsky 2009:p. 27).
 - "No one is so deluded as to believe that there is a mind-independent object corresponding to the internal syllable [ba], some construction from motion of molecules perhaps, which is selected when I say [ba] and when you hear it" (Chomsky 2015:p.126).

4 Segments, natural classes and underspecification

The point of this section is to clarify the long recognized problem of referring to a relatively unmarked representation R (a segment, say) without referring to all representations that contain all of the information in R and more, the representations that are subsumed by R .⁴ Bale et al. (2014) found a principled solution to this problem in the domain of changes of features that occur within a segment. I explain the issue in the context of a fairly non-controversial view of features, segments and natural classes. If this logic is obvious, this section can be skipped—in the next section we see the insufficiency of the solution in the case of full *segment* insertion and deletion rules.

Bale et al. (2014) and subsequent work (e.g. Bale et al. 2020, Volenec and Reiss 2020, Reiss 2021) sometimes referred to as ‘Substance Free Logical Phonology’ (SFLP) make a point of distinguishing the set-theoretic types of different parts of a rule statement. In the SFLP formalism, as in older models, the target and environment of rules refer to natural classes of segments—this constitutes part of a theory of rule syntax. However, the structural change of a rule refers merely to the set of features that are changed in some manner. Consider the informal rule in (8).

$$(8) \quad \begin{array}{c} \text{o,e} \\ \text{Target} \end{array} \rightarrow \begin{array}{c} \text{u,i} \\ \text{Change} \end{array} / \text{ — } \begin{array}{c} \text{m,n,ŋ} \\ \text{Environment} \end{array}$$

Translated into *SPE*-style feature notation this rule might be rewritten as in (9) with all feature specifications in square brackets, as in much of the phonological literature.

$$(9) \quad \begin{array}{c} [-\text{Low}] \\ \text{Target} \end{array} \rightarrow \begin{array}{c} [+H\text{IGH}] \\ \text{Change} \end{array} / \text{ — } \begin{array}{c} [+N\text{ASAL}] \\ \text{Environment} \end{array}$$

⁴This issue is recognized explicitly by Lees (1961) and Lightner (1971:p. 236), for example. Kiparsky’s (1985:p. 98) discussion of coronal underspecification briefly notes this problem too, stating that it is necessary “to work out some way of referring to unmarked segments” when representations are not fully specified. His solution was to put an ‘x’ on association lines that lead to nothing, but this raises the question of exactly what specification is absent: to *which* ‘nothing’ do the lines (not) lead? Archangeli (1988) offers a similarly unsatisfactory solution. These problems and a proposed solution are discussed at length in Reiss (2003). The solution adopted in this paper is derived from Bale et al. (2014). Hayes’ 1986:p. 331 Linking Constraint—“Association lines in structural descriptions are interpreted as exhaustive”—is an attempt to deal with the problem in the context of autosegmental representations. The necessity of distinguishing feature-filling from feature-changing processes is intimately related to this problem.

The SFLP literature adopts the practice of using square brackets for the natural classes in a rule's target and environment, but using normal set theoretic curly brackets for the set of features specified in the rule's change, as in (10).⁵

$$(10) \quad \begin{array}{ccc} [-\text{Low}] & \rightarrow & \{+\text{HIGH}\} \\ \text{Target (SD)} & \text{Change} & \text{Environment} \end{array} \quad / \text{---} \quad [+ \text{NASAL}]$$

SFLP provides a set theoretic interpretation of the square brackets, to which we now turn.

For the kinds of phenomena examined thus far in SFLP, it is sufficient to treat segments as sets of valued features. Therefore, individual segments, just like structural changes, are denoted in regular set-theory curly brackets, as in (11) (assuming that only the features listed here are relevant).

$$(11) \quad \text{Segments /i/ and /e/ as sets of features}$$

$$/i/ = \left\{ \begin{array}{c} +\text{SYLLABIC} \\ -\text{BACK} \\ -\text{ROUND} \\ +\text{HIGH} \\ -\text{LOW} \\ +\text{ATR} \end{array} \right\} \quad /e/ = \left\{ \begin{array}{c} +\text{SYLLABIC} \\ -\text{BACK} \\ -\text{ROUND} \\ -\text{HIGH} \\ -\text{LOW} \\ +\text{ATR} \end{array} \right\}$$

A set like (12) is also a legitimate representation of a segment, denoting a vowel that lacks a value for the feature HIGH, but is otherwise identical to /i/ and /e/.

$$(12) \quad \text{The vowel /E/ as a set of features}$$

$$/E/ = \left\{ \begin{array}{c} +\text{SYLLABIC} \\ -\text{BACK} \\ -\text{ROUND} \\ -\text{LOW} \\ +\text{ATR} \end{array} \right\}$$

In other words, SFLP allows for underspecification: there is no requirement that each segment be *complete*, specified for all features.

The representation in (12) must be distinguished from the representation in (13) which is a 'partial description'. SFLP uses square brackets to denote natural classes, so (13) is an intensional characterization of a set of segments, where each segment is a set of features.

$$(13) \quad \text{The natural class containing /i, e, E/}$$

$$\left[\begin{array}{c} +\text{SYLLABIC} \\ -\text{BACK} \\ -\text{ROUND} \\ -\text{LOW} \\ +\text{ATR} \end{array} \right]$$

⁵Of course, this use of curly brackets has no relationship to the traditional use in generative phonology to denote disjunctions in rules—SFLP does not allow disjunctive statements in rules.

The square brackets are used to avoid the unwieldy expression in (14).

$$(14) \quad X = \{x : x \supseteq \{+\text{SYLLABIC}, -\text{BACK}, -\text{ROUND}, -\text{LOW}, +\text{ATR}\}\}$$

This is just the set of segments whose members are all and only the segments that are supersets of the set of features $\{+\text{SYLLABIC}, -\text{BACK}, -\text{ROUND}, -\text{LOW}, +\text{ATR}\}$.

With the representations of /i, e, E/ above, we now see that it is possible to denote a natural class consisting of a single member segment, say /i/, as in the left side of (15) which is not the same as the ‘direct’ representation of the segment /i/ on the right side, which uses normal set brackets.

$$(15) \quad \text{The natural class } \{i\} \text{ is not the same as the segment } /i/$$

$$\left[\begin{array}{c} +\text{SYLLABIC} \\ -\text{BACK} \\ -\text{ROUND} \\ +\text{HIGH} \\ -\text{LOW} \\ +\text{ATR} \end{array} \right] \neq \left\{ \begin{array}{c} +\text{SYLLABIC} \\ -\text{BACK} \\ -\text{ROUND} \\ +\text{HIGH} \\ -\text{LOW} \\ +\text{ATR} \end{array} \right\}$$

Of course, we can also represent the natural class that contains only the segment /e/ by changing +HIGH to –HIGH on the left side of (15).

In contrast, it is *not* possible to denote a natural class that contains only the segment /E/. This is because the feature set corresponding to /E/ is a subset of the sets corresponding to each of /i/ and /e/, so these latter two are supersets of /E/. In general, there is no way to refer to a natural class containing a segment p that is underspecified relative to another segment q , that is, where $p \subseteq q$. The segment /E/ contains all the features of the natural class representation in (13), but /i/ and /e/ also contain all these features. David Ta-Chun Shen (Bale et al. 2020) noted that a nice outcome of this formalization of segments and natural classes is that it makes it unnecessary to have a feature +SEGMENT to refer to the natural class of all segments in a language. This class can be referred to as just $[\]$, because basic set theory tells us that every set (every segment) is a superset of the empty set (see below).

The focus of much work in the SFLP corpus has been the development and application of a mechanism to overcome this problem: How can feature-filling processes be formulated if it is not possible to refer to relatively underspecified segments? The SFLP solution has involved building into the rules that make changes in the feature content of segments a unification operation ‘ $\sqcup$ ’ that takes two sets as its arguments. The unification of a segment (set) and the set of features corresponding to the structural change is equivalent to the set theoretic *union* of the two sets, but only if

the union of the two sets is consistent, if it contains no conflicting feature values, like +F and −F. If the union of the two sets is not consistent, unification *fails* (or equivalently, its output is *undefined*). Unlike union, which always yields an output set for any two input sets, and is called a total operation, unification is called a partial operation, a property that SFLP exploits.

Application of a rule that unifies the segment /i/ with the set {+HIGH} will apply vacuously, because the set corresponding to /i/ already contains the element +HIGH: so $/i/ \sqcup \{+HIGH\} = /i/$. The unification succeeds since $/i/ \cup \{+HIGH\}$ yields a consistent set of features, the segment /i/. The same rule applied to the segment /e/, which is specified as −HIGH, will result in unification failure, because of the SFLP assumption that segments must be consistent, they cannot contain both +F and −F, for any feature. So $/e/ \sqcup \{+HIGH\}$ is **undefined** as a logical operation, because $/e/ \cup \{+HIGH\}$ is not consistent. However, the rule semantics of SFLP determines that when unification fails, the target segment, the first argument of the unification operator, remains unchanged. Therefore, in the application of a *phonological rule*, the unification of /e/ with {+HIGH} yields /e/, another kind of vacuous rule application.

Finally, the unification of /E/ with {+HIGH} is equivalent to the union of the two sets, and thus yields /i/: $/E/ \sqcup \{+HIGH\} = /i/$. The effect is that the rule *appears* to only have targeted /E/, but in fact the rule’s application is vacuous for /i/ and /e/ (for different reasons) and only application to /E/ is non-vacuous.⁶

Work such as Reiss (2021) provides copious applications of this mechanism, along with a second type of rule involving set subtraction, denoted here with ‘\’, which can combine with unification rules to yield feature-changing processes: subtraction can remove features and unification can

⁶This solution based on unification failure, first suggested by Bale et al. (2014), thus addresses the concern of Lees (1961):

Thus, we must assume that any rule which applies to a column of features like /B/ also at the same time applies to every other type of column which contains that same combination of features, such as /p/ and /b/. This is tantamount to imposing the constraint on phonological features that they never be required to identify unspecified, or zero, features. To the best of our present knowledge, there seems to be no other reasonable way to prevent the awkward consequences mentioned above.

To return to Turkish this decision means that the grammar is incapable of distinguishing native vowel-harmonic morphemes from borrowed non-vowel-harmonic morphemes simply be the presence of the archiphoneme /E/ in the former versus /e/ or /a/ in the latter.

add them. Set subtraction $A \setminus B$ yields a set C that contains all the members of A that are not in B . For example, $\{a, b, c\} \setminus \{a, x, y\} = \{b, c\}$.

The combination of unification and subtraction can cover a wide range of phonological patterns, but the solution offered by unification to the problem of referring to underspecified segments does not extend to processes that look beyond the feature sets that comprise segments. In other kinds of processes, such as *segment* insertion and deletion, the apparent targeting of underspecified segments afforded by unification is not available.

5 Insertion and deletion of segments (not features)

Previous SLFP literature has explicitly focused on intrasegmental changes, but has also recognized the necessity of looking beyond the sets of features that make up segments. In particular, it is necessary to consider the timing slots to which sets of features are associated in order to study the nature of segment insertion and deletion rules. The SFLP literature uses the operation of set subtraction to remove feature specifications from segments. It is crucial to note that deletion of a segment is *not* the same as deletion of all the features associated with a segment. The empty set contains nothing, but it is not nothing, so an X-slot associated to an empty set is different both from an X-slot unassociated to a set and from the absence of an X-slot altogether.⁷ In light of such considerations, the simple set representations in (11) have to be enhanced at least to something like (16). These structures can be further incorporated into syllable and foot structures as well.

(16) Segments as sets of features associated to X-slots

$/i/ =$	X $\downarrow$ $\left\{ \begin{array}{c} +\text{SYLLABIC} \\ -\text{BACK} \\ -\text{ROUND} \\ +\text{HIGH} \\ -\text{LOW} \\ +\text{ATR} \end{array} \right\}$	$/e/ =$	X $\downarrow$ $\left\{ \begin{array}{c} +\text{SYLLABIC} \\ -\text{BACK} \\ -\text{ROUND} \\ -\text{HIGH} \\ -\text{LOW} \\ +\text{ATR} \end{array} \right\}$
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In other words, the fact that SFLP has not yet addressed such higher-order structures should not be taken to mean that the approach is incompatible with them. For example, the reasoning outlined in the previous section

⁷This conflation of the empty set and nothing is at the root of Anderson's notational problems discussed above.

concerning natural classes can be applied to categories like, say, nasals in coda position, where both feature sets and syllable structure are relevant.

In order to avoid Anderson’s dual use of the symbol ‘ $\emptyset$ ’, discussed above, I adopt the empty *segment* symbol ϵ used in the SFLP literature (see Bale and Reiss (2018) for discussion, including a sketch of the semantics of rules with ϵ). With this symbol, we can formulate insertion and deletion rules, rules containing expressions like ‘ $\epsilon \mapsto p$ ’ which inserts a p or ‘ $p \mapsto \epsilon$ ’ which deletes a p .⁸

Actual deletion rules target classes of segments, such as nasals or high vowels, just like unification rules do. So, a rule deleting nasals word finally will have to be formulated as something like (17).

(17) Rule deleting word-final nasals

$$\begin{array}{c} \text{X} \\ | \\ [+\text{NASAL}] \end{array} \mapsto \epsilon \quad / ___ \%$$

The target of this rule is the set of X-slots in word-final position associated to a set of features that is a superset of $\{+\text{NASAL}\}$. In other words, the square brackets indicate that the whole expression in target position represents a partial description, a natural class.

We will now see that the unification-based solution proposed in SFLP that allows the formulation of rules that appear to target only relatively underspecified segments is not available in the case of segment insertion and deletion rules. The argument is a purely logical one, but it has ramifications for widespread claims about phonological systems.

6 Deletion cannot target ‘weak’ / $\emptyset$ /

The symbol / $\emptyset$ / is used in various ways in the phonological literature. Most straightforwardly, it is an IPA symbol used to denote a mid, central, unrounded, lax vowel, perhaps with the featural make up in (18)⁹.

(18) Schwa as a set of features

⁸For informal rules like (8) I used the traditional ‘ $\rightarrow$ ’ symbol of rule based phonology. The SFLP program deconstructs this operator into the unification and subtraction operators discussed above, and the current paper adds a third operator ‘ $\mapsto$ ’ for segment insertion and deletion.

⁹These features suffice for our main point. We leave aside orthogonal questions such as whether a ‘normal’ / $\emptyset$ / is specified as $-\text{NASAL}$ or even $-\text{LATERAL}$.

$$/\partial/ = \left\{ \begin{array}{l} +\text{SYLLABIC} \\ +\text{BACK} \\ -\text{ROUND} \\ -\text{HIGH} \\ -\text{ATR} \\ -\text{LOW} \end{array} \right\}$$

We are not concerned with this notion of $/\partial/$ as a ‘normal’ vowel. This kind of schwa occurs in many languages, where it behaves like other vowels, for example, occurring in both stressed and unstressed syllables.

However, the same IPA symbol is also used to refer to a vowel that ‘varies’ with the absence of a vowel. Sometimes, this variation refers to supposed ‘free variation’ depending on speech rate, register or other paralinguistic or sociolinguistic factors, and sometimes it refers to actual phonologically triggered conditions involving syllable structure, stress and the like. This last use of $/\partial/$ is the one that concerns us—an underlying ‘weak’ vowel that putatively alternates with the absence of a vowel by virtue of phonological computations, sometimes with additional morphological conditioning. Examples abound but for concreteness, we can cite (Booij 1977:p. 81) who has a rule deleting $/\partial/$ in Dutch before a certain class of suffixes beginning with a vowel. This rule is adopted by (Scholten 1987:p. 56) who explicitly assumes that Dutch $/\partial/$ is completely underspecified, containing no features at all.

Typically, but not always, the phonetics of this vowel corresponds, when pronounced, more or less to that of the IPA symbol, as well as to intuitions of how the feature bundle in (18) would be pronounced.¹⁰

Adopting the SFLP notation, a non-controversial representation of a ‘weak’ $/\partial/$ might be something like (19), where the lack of specification for other vowel features is supposed to explain this vowel’s tendency to disappear.¹¹

$$(19) \text{ ‘Weak’ schwa: } /\partial/ = \{ +\text{SYLLABIC} \}$$

We can now attempt to formulate a rule to delete such a schwa. Of course, if we want to maintain the SFLP assumption that there is a logically coherent

¹⁰A well-known case of a ‘ ∂ ’ that alternates with zero, but does not have a pronunciation corresponding in an obvious way to (18) is the ‘French schwa’ that Anderson discussed, which, when pronounced, is usually realized as a mid, front, rounded vowel, something like an IPA $[\text{œ}]$. Calling this vowel ‘schwa’ raises a number of questions orthogonal to our concerns here, for example, the question of whether the transduction from features to the output of the phonetic module is invariant or language-specific. We don’t pursue these issues here.

¹¹Of course, Anderson does not need the feature $+\text{SYLLABIC}$ because his segment $\emptyset$ is associated with a nucleus. The question arises of how one tells that there is an association with ‘nothing’. The answer is given by our understanding of $\emptyset$ as the empty set, but Anderson does not do this—for him the symbol denote ‘nothing’.

syntax of rules, the target of the rule must be expressed as a natural class. The set of features corresponding to ‘weak’ /ə/ contains, by hypothesis, only +SYLLABIC, so we might be tempted to formulate a rule like (20). The environment is specified in generic fashion as P_Q .

(20) Rule intended to delete only ‘weak’ /ə/

$$\begin{array}{c} X \\ | \\ [+\text{SYLLABIC}] \end{array} \mapsto \epsilon \quad /P \quad _ \quad Q$$

Of course this rule will not work—it will overapply and delete all vowels in the environment P_Q because every vowel is a superset of {+SYLLABIC}. For a less extreme case, consider a rule that would delete the underspecified vowel /E/ discussed above. It is not possible to formulate such a rule without having it also delete the vowels /i/ and /e/.¹²

7 Delete the rich

It is of course possible to delete a *richly* specified vowel, as in (21), by defining the target as a natural class with just one member.

(21) Rule deleting fully specified /i/

$$\begin{array}{c} X \\ | \\ \left[\begin{array}{l} +\text{SYLLABIC} \\ -\text{BACK} \\ -\text{ROUND} \\ +\text{HIGH} \\ -\text{LOW} \\ +\text{ATR} \end{array} \right] \end{array} \mapsto \epsilon \quad /P \quad _ \quad Q$$

This vowel /i/ in this rule is an arbitrary choice, of course—a fully specified ‘strong’ schwa could also be deleted by reference to the natural class containing just this one vowel as in (22).

(22) Rule deleting fully specified ‘strong’ /ə/

¹²I am abstracting away from factors such as stress. If underlying or derived ə is the only unstressed vowel, and it gets deleted, then we can attribute the deletion to lack of stress. We don’t need the additional hypothesis that the vowel is underspecified in addition to being unstressed in order to explain its deletion. As phonetician John Ohala was fond of saying, in science, having two explanations is not better than having one.

$$\begin{array}{c} X \\ | \\ \left[\begin{array}{c} +\text{SYLLABIC} \\ +\text{BACK} \\ -\text{ROUND} \\ -\text{HIGH} \\ -\text{LOW} \\ -\text{ATR} \end{array} \right] \end{array} \mapsto \epsilon \quad /P \quad ___\quad Q$$

It is also possible to delete members of a particular natural class containing several vowels as in (23), which deletes just the high vowels in a language.

(23) Rule deleting all high vowels

$$\begin{array}{c} X \\ | \\ \left[\begin{array}{c} +\text{SYLLABIC} \\ +\text{HIGH} \end{array} \right] \end{array} \mapsto \epsilon \quad /P \quad ___\quad Q$$

This rule deletes every vowel that is a superset of the set $\{+\text{SYLLABIC}, +\text{HIGH}\}$. A rule deleting strong /ə/ along with some other vowels, such as all the non-low vowels in a language, can also be formulated by replacing +HIGH in (23) with –Low.

I have argued that the deletion of what surface as ‘ə-colored’ vowels cannot be due to their lack of structure, for purely set-theoretic reasons. Of course, one could propose a different logic for phonology in which the absence of structure can be referenced and lead to deletion. In other words, I accept that the conclusion I have reached is dependent on my premise of a binary feature system allowing underspecification, but without *some* premises, no conclusions can ever be reached. A privative feature system will lead to other conclusions, for example.

However, I have also suggested that the widespread equation of impoverished representational content and a disposition to be deleted is conceptually suspect, so the pursuit of an alternative logic may not be advisable. In section 8, I show that ‘weak’ vowels can be selectively *inserted*. Then in section 9, I provide empirical evidence against, or rather highlight the apparent lack of evidence for, weak schwa deletion as an attested process.

8 Insert the poor

In contrast to the impossibility of targeting for deletion relatively underspecified segments with impoverished representations, it is possible to target them for insertion. Thus, an inserted [ə], a vowel that does not appear in the underlying representation of a derivation, can be the ‘weak’ vowel

of (19), whether it surfaces with the phonetics of an IPA [ə], copies features from neighboring vowels, or receive specification by other means.

In order to understand why insertion of weak [ə] is possible, we need to go back to the type-theoretic analysis of rules. Recall that the target and environment of a rule contain reference to partial descriptions that define natural classes. This is not the case for the structural change. The structural change of a rule like (10) is just a set of features. It is not a partial description that intensionally defines a natural class. In the case of a segment insertion rule, the inserted content is an X-slot with an associated set of valued features. So a fully specified vowel can be inserted by a rule like (24):

(24) Rule inserting fully specified /i/

$$\epsilon \mapsto \left\{ \begin{array}{c} \text{X} \\ \text{+SYLLABIC} \\ \text{-BACK} \\ \text{-ROUND} \\ \text{+HIGH} \\ \text{-LOW} \\ \text{+ATR} \end{array} \right\} /P \quad _\quad Q$$

Note that the inserted material is an X-slot associated to a *set* of features, shown in curly brackets, unlike the case of the target in rule (21), where square brackets define a target natural class that happens to contain a single member.

We can now provide a rule that inserts a weak /ə/ in (25).

(25) Rule inserting weak ə

$$\epsilon \mapsto \left\{ \begin{array}{c} \text{X} \\ \text{+SYLLABIC} \end{array} \right\} /P \quad _\quad Q$$

To push further, let's return to the point mentioned above concerning the use of the expression [] to refer to the natural class of all segments. As suggested, we can formulate a rule deleting all segments in a given environment, as in (26).

(26) Rule deleting all segments

$$\left[\begin{array}{c} \text{X} \\ \text{ } \end{array} \right] \mapsto \epsilon /P \quad _\quad Q$$

The target of this rule is the class of X-slots associated with any set of features that is a superset of the empty set, so any segment whatsoever. In contrast, consider a rule like (27).

(27) Rule inserting an empty set of features

$$\epsilon \mapsto \left\{ \begin{array}{c} X \\ | \\ \end{array} \right\} / P \quad \text{---} \quad Q$$

This rule inserts an X-slot associated with the empty set, a segment that contains no features whatsoever, not even the +SYLLABIC specification in (25). This inserted segment can receive specifications from other segments in the context, as in the behaviour of epenthetic copy vowels, or the addition of a timing slot that may take on vocalic or consonantal features. It is perhaps useful to reiterate here the point made in fn. 2, that the absence of features in the set associated with a timing slot does not denote the absence of a segment. So if weak schwa is truly devoid of any specified features, then (27) is the correct formulation of a rule to insert such a segment.

9 Some surprising facts

I assume that even readers sympathetic to the rule-based framework I adopt here will find repugnant my conclusion concerning the impossibility of deleting ‘weak’ schwa, and will object that although my logic may be valid, it derives a from a set of unjustified premises.¹³ In anticipation of such objections, I suggest that the premises are merely a formalization of standard views in the field, such as the idea that natural classes are defined by shared sets of valued features, and that such results are exactly what one expects as soon as one decides to be explicit. Appreciation of this attitude is an “important trait” according to Zenon Pylyshyn (1984:p. xxii):

¹³Bjoern Koehnlein (p.c.) points out that in Optimality Theory grammars, it would appear to be predicted that weak segments are easier to delete because their deletion violates fewer Max-Feature constraints than the deletion of richer segments. (On the other hand, weak segments should be *avored* on the surface since they violate fewer *STRUCTURE constraints. Another issue is that much of the OT literature rejects underspecification, ignoring the arguments of Inkelas (1995) that underspecification is necessary, even in OT, in particular her point that the existence of bad arguments for underspecification should not lead to the conclusion that underspecification is bad—she provides good arguments that it is necessary.) As we will see below, the empirical evidence for schwa deletion is not at all robust, and so Koehnlein's proposed Optimality Theory prediction, while consistent with phonologists' intuitions, may be empirically unsupported.

[I]f you believe P , and you believe that P entails Q , then even if Q seems more than a little odd, you have some intellectual obligation to take seriously the possibility that Q may be true, nonetheless. That is what it means to take ideas seriously.

A similar sentiment is suggested by Chomsky 1957:p. 5 in the preface to *Syntactic Structures*:

The search for rigorous formulation in linguistics has a much more serious motivation than mere concern for logical niceties or the desire to purify well-established methods of linguistic analysis. Precisely constructed models for linguistic structure can play an important role ...[A] formalized theory may automatically provide solutions for many problems other than those for which it was explicitly designed. Obscure and intuition-bound notions can neither lead to absurd conclusions nor provide new and correct ones, and hence they fail to be useful in two important respects. I think that some of those linguists who have questioned the value of precise and technical development of linguistic theory may have failed to recognize the productive potential in the method of rigorously stating a proposed theory and applying it strictly to linguistic material with no attempt to avoid unacceptable conclusions by *ad hoc* adjustments or loose formulation.

We have seen how the set-theoretic formalization of segments and natural classes leads to a result about segment deletion that initially strikes us as an “unacceptable conclusion” and “more than a little odd.” However, we are under “some intellectual obligation to take seriously the possibility that [it] may be true, nonetheless.”

Instead of being satisfied with these philosophical musings, let’s consider whether the impossibility of deleting ‘weak’ schwa really warrants concern. It is true that Flemming (2009) says that “Schwa is often singled out by deletion processes, e.g. in Dutch (Booij 1985), English (Hooper 1978), French (Dell 1973) and Hindi (Ohala 1983),” however it is not at all clear that these cases concern a ‘weak’, highly underspecified vowel. For Dutch, van Oostendorp (1995) has reanalyzed schwa deletion as insertion, and this analysis was adapted for Frisian as well by de Haan (1999) who explains “My alternative is based on a proposal of van Oostendorp (1995), who suggests that phonologically regular schwa-deletion should be accounted for lexically: cases of schwa-deletion are derived by base-generating schwa-less syllables, followed by schwa-epenthesis.”

Hooper's discussion focusses on speech rate and register variation, not phonological context, and is skewed towards explaining diachronic developments. French 'schwa', as mentioned above, is well-known to be a front, round vowel phonetically, and thus unlikely to be featurally impoverished.¹⁴ Discussions of putative schwa deletion in Hindi and other Indo-Aryan languages tend to conflate synchronic and diachronic discussion, and furthermore, the vowel in question tends not to be pronounced like an IPA [ə]. In his section on schwa-zero alternations, Silverman (2011) says that "According to Ohala (1999), the vowel in alternation with zero is actually slightly lower than schwa: [a]. It is transcribed [ə] in Ohala (1983), and herein as well, for the sake of internal consistency." So, this again, like the French 'schwa' is not an articulatorily neutral vowel. The vowel is *called* schwa *because* it alternates with zero. The only other schwa-zero alternation Silverman discusses is what has been analyzed as an insertion rule in Chukchee by Kenstowicz (1994).¹⁵

These examples appear to be typical, and the empirical evidence for weak schwa deletion as a phonological process thus appears to be pretty thin. In some places, the existence of weak schwa deletion is explicitly taken as a corollary of the possibility for insertion of schwa (in the sense of a highly underspecified vowel). This kind of reasoning is explicitly laid out by Scholten (1987:p. 56):

Given that [ə] is represented in the underlying representation as a completely unspecified timing unit (in addition to Piggott and Singh (1985), see also ter Mors (1985), who argues for this representation for Klamath epenthesis and Archangeli (1988) who uses a similar treatment for Yawelmani), then simple insertion of a timing unit in any position ought to be realized as an epenthetic vowel. Deletion of schwa is also quite straightforward since it involves just the deletion of a timing unit under certain conditions.

¹⁴Absent an ad hoc model of language-specific phonetic implementation that makes phonology unnecessary. Hualde (1991) provides an excellent discussion of the necessity to distinguish underspecified vowels from default vowels.

¹⁵Of course, we don't demonstrate that schwa deletion is real by calling every lexical vowel that deletes 'schwa', regardless of its phonetics. Furthermore, we don't demonstrate that 'weak' (impoverished) schwa deletes just by showing that a vowel pronounced like IPA [ə] deletes. My claim is that it is only if a phonetically schwa-like vowel corresponds to a set of features that is not a subset of the set corresponding to any other vowel in a language, that that schwa-like vowel can be targeted by a deletion rule, to the exclusion of any other vowel.

It turns out that the three sources cited by Scholten all focus on vowel epenthesis, not deletion. The error of reasoning is a failure to recognize the lack of symmetry between insertion and deletion: in insertion, the rule’s change is a segment, the particular segment that gets inserted; but in deletion, the rule’s target is a natural class, the set of segments that get deleted. This contrast was made clear in the comparison of (26) vs. (27). For a natural class, TDH tells us that there is an inverse relation between the cardinality of a set of features used in defining the class and the size of the class: more features in the intensional characterization yields a smaller class of segments. For example, the size of the set of voiced stops is smaller (or equal to) the size of the set of stops in a language. Failure to appreciate TDH has led to confusion in several domains of phonology, as discussed in Hale and Reiss (2003) and Gorman and Reiss (2023), *inter alia*. Scholten’s parallel fails because, like other discussions of weak schwa deletion, it conflicts with TDH.

Another source of the chimera of schwa deletion appears to be due to an extrapolation from fairly common processes of vowel reduction (often to a phonetic schwa) to the idea that the next logical step from reduction, as deletion of *features*, is deletion of a whole segment. Removal of features in SFLP is effected by rules based on set subtraction, so a rule that ‘reduced’, say, an /i/ to a vowel closer to a weak schwa can be expressed trivially (see Reiss 2021 for more examples), as in (28).

$$(28) \quad \text{Rule deleting } \textit{features} \text{ from } /i/$$

$$\begin{array}{c} \text{X} \\ \downarrow \\ \left[\begin{array}{l} +\text{SYLLABIC} \\ -\text{BACK} \\ -\text{ROUND} \\ +\text{HIGH} \\ -\text{LOW} \\ +\text{ATR} \end{array} \right] \setminus \left\{ \begin{array}{l} -\text{BACK} \\ -\text{ROUND} \\ +\text{HIGH} \\ +\text{ATR} \end{array} \right\} /P \quad \text{---} \quad Q \end{array}$$

The target class of this rule has only a single member, /i/. Recall that set subtraction, $A - B$ yields a set C whose members are the members of A that are not in B : for example, $\{x, y, z\} - \{x, a\} = \{y, z\}$. Thus the effect of this rule is to turn an /i/ into a vowel whose only features are $\{+\text{SYLLABIC}, -\text{Low}\}$.

In light of our proposed representation of segments as *sets* of features associated to an X-slot, the source of confusion becomes clear. Even if a segment is ‘reduced’ by deleting *all* its features, the segment itself remains, and not just as an X-slot. Remember that an empty set is not nothing—it is an empty set, which in our model of segment structure, is associated to

an X-slot as in (29a). This differs from an unassociated X-slot (29b), and it differs from the nothingness in (29c).¹⁶

(29) A very ‘weak’ segment vs. an X-slot vs. nothing

a.	X { }	b.	X	c.	
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This thing in (29a) can be inserted—either with a specification for +SYLLABIC as in the representation in (25), above, or even completely underspecified, as in (30=27).¹⁷

(30) Rule inserting weak ə

$$\epsilon \mapsto \left\{ \begin{array}{c} X \\ | \end{array} \right\} / P \quad _ \quad Q$$

Such a vowel (or segment, more generally) could exit the phonology, totally unspecified, as what Crosswhite (2004) calls a “targetless” vowel, using a common expression for the idea of lacking featural specification. Another possibility is that it would receive some or all features by context-free redundancy rules, as Anderson proposes for French schwa. Finally, the vowel might receive some of its features from the phonological context, as in the case of so-called ‘copy epenthesis’ vowels that vary according the environment of insertion, sometimes copying all features from a vowel in a neighboring syllable (see Stanton and Zukoff (2018) for examples and OT analyses). Another behaviour of highly underspecified vowels is the extreme coarticulation effects seen in vertical vowel systems, such as Marshallese (see Bender 1968, Choi 1992, Hale and Reiss 2008), where vowels are specified only for HIGH and Low, and remain underspecified

¹⁶Of course, we regularly use the ‘nothing’ in (29c) to mean ‘anything’, as in a rule of the form $a \rightarrow b / c _$, where specifying nothing after the ‘ $_$ ’ symbol means that anything at all can appear there—there are no constraints on what appears there.

¹⁷The unassociated X-slot, as in (29b), is probably not a possible representation, despite its common occurrence in the literature. Such X-slots are typically used to provide an anchor for the features of a neighboring segment, for example, to express vowel length. However, since (29b) subsumes (is more general) than (29a) as well as any more specified segment, it is clear that there is no way to refer to just (29b). What is commonly represented as (29b) in the literature should be interpreted, instead, as (29a)—an X-slot can only be inserted with an associated feature set (potentially the empty set). This set can then be filled, via unification, with valued features potentially copied from other segments, yielding processes that go by various names like ‘vowel copy’ and ‘lengthening’.

even in the output of the grammar for features like BACK and ROUND, but do receive (non-featural) coarticulatory effects from flanking consonants for those features.

Let's finish with a discussion of schwa that is notable for its clarity in representing what seem to be the shared intuitions of the phonological community, a survey entitled 'Schwa in phonological theory':

If schwa is very simple [i.e. underspecified—c.r.], there are reasons to assume that this vowel is particularly eligible for both epenthesis and deletion. In the case of epenthesis, it is the most economical choice to make if we have to insert a vowel: no material needs to be epenthesized beyond the feature [–consonantal] [or alternatively +SYLLABIC, as in this paper—c.r.]. In the case of deletion, it is simpler to delete only the specification [–consonantal] than to delete this feature alongside others (which is needed in case of the deletion of a full vowel). It is thus predicted that if a language (i) allows schwa on the surface (of a given subphonology) and (ii) has a vowel-zero alternation (within that same subphonology), schwa will be among the alternating vowels. I have no evidence that would falsify this prediction; I know of no study which tries to systematically verify it either. (van Oostendorp 2003:p. 435)

This passage reiterates the idea that deleting a segment is just an extreme case of deleting all of its features, but under our representation of segments as X-slots linked to *sets* of features, this idea does not hold, as made clear by the contrasts in (29). The passage also suggests that deleting a single feature is obviously simpler than deleting several features.

Interestingly, Bale et al. (2020) argue that, although unification rules are restricted to unifying segments with singleton sets of features (somewhat along the lines of Chomsky's 1967 Principle 6: "Two successive lines of a derivation can differ by at most feature specification") it has to be the case—it follows from the logic of the TDH Principle—that subtraction rules can, and typically must, delete multiple features at once. So, even the intuitive simplicity of deleting less rather than more may be an illusion. The quoted passage above then makes the somewhat vague prediction that 'weak' schwa should be among a set of vowels that participate in any vowel-zero alternation due to deletion. In contrast, we make the explicit claim that if a 'weak' vowel deletes, then any other segment whose feature set is a superset of those belonging to the weak vowel will also delete. Finally, although van Oostendorp's logic may work for some versions of Optimal-

ity Theory, it is clearly at odds with the explicit logic offered by SFLP in this paper:

[D]eletion of schwa is less costly than deletion of other vowels, due to the fact that the schwa literally is a substructure of all the other vowels. If it is allowed to delete a full vowel in a certain context, this implies that it is allowed to delete a root node in that context; hence, also schwa will be deletable in that context. (van Oostendorp 2003:p. 438)

In SFLP, the logic is inverted because of TDH, the less information used to intensionally define a natural class, the larger the class: deletion of weak schwa entails the deletion of full vowels, not vice versa, as we have seen. Yes, a weak schwa corresponds to a “substructure” of other vowels, but the natural class defined with such a substructure necessarily includes the other vowels—this is the logic of partial descriptions at the base of the notion of natural classes.

Despite the longstanding assumption in the field that weak schwa deletes easily, it is noteworthy that van Oostendorp’s own (1995) analysis of Dutch and Vaux’s (1998) analysis of Armenian both posit epenthesis rather than syncope. Vaux, for example, concludes that “it is preferable to assume that Armenian does not in fact possess a schwa in its phonemic inventory, and that all schwas which appear in surface representations result from a process of epenthesis” (p. 70). So, in the end, the empirical facts, the lack of clear evidence for the deletion of weak schwa, appear to align fairly well with the predictions of our explicit formal theory: ‘weak’ [ə] deletion does not exist as a phonological process. This conclusion seems “more than a little odd”, but we have some obligation to take it seriously, nonetheless.

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